Higher National Diploma in Engineering (Electrical)
4th Year, First Semester Examination – 2020
EE 3214 – Construction and Protection of Electrical systems

Instructions for Candidates:	No. of questions	: 06
Answer **only five** questions.	No. of pages	: 04
each questions carry equal marks	Time	: **03 hours**

01.

a). A single phase 20 kVA, 480/120V, 60 Hz single phase transformer has an equivalent impedance of 0.0525<78.13 degrees ohms referred to the Low Voltage (LV) winding side.

i. Determine equivalent impedance referred to the High Voltage (HV) winding side.
ii. Determine the per unit transformer impedance referred to the LV winding and the HV winding

(10 Marks)

b) Draw the single-phase equivalent per unit reactance diagram for the circuit shown in the fig Q1, Working on a system base of 100 MVA and a base voltage of 115 kV in the load circuit and reactance of the generator is neglected (X_g =0) .

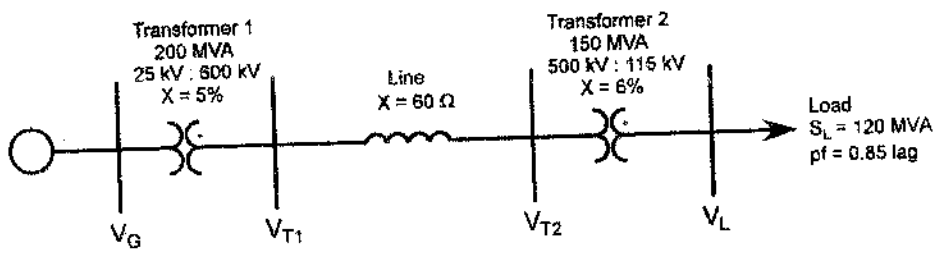


Fig Q1

(10 Marks)

02.

a) A small generating station has two alternators of 4000 kVA and 5000 kVA and percentage reactance of 0.08 p.u and 0.09 p.u respectively. The circuit breakers have a rupturing capacity of 120 MVA. It is desired to extend the system by a supply from the grid via a transformer of 7500 kVA and 8% reactance as shown in the fig Q2.a Assume the bus voltage = 3300 V

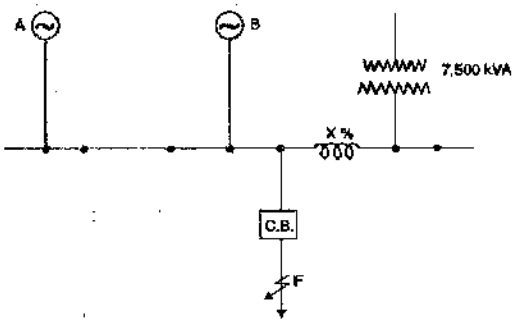


Fig Q2.a

i. What is the function of the Current limiting reactor used in the bus bar section? (3 Marks)

ii. Find the reactance (X in Ω) of the reactor connected in the bus bar, if a symmetrical short-circuit occurs on an outgoing feeder connected to it. (7 Marks)

b) Two generators G1 and G2 are rated 15MVA, 11KV,0.1p.u reactance and 10MVA, 11KV,0.1p.u reactance respectively. The generators are connected to a step up transformer of 11/66kV, 0.06 p.u reactance as shown in fig.Q2.b Calculate current generated in each generator when a three-phase fault occurs on the high voltage side of the transformer.

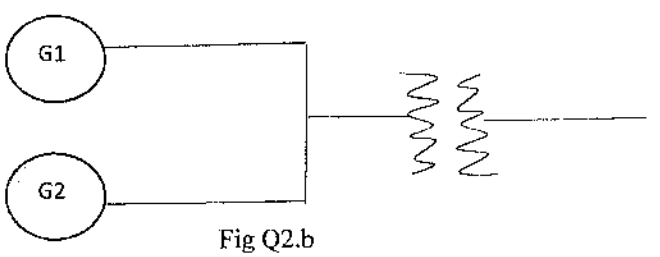


Fig Q2.b

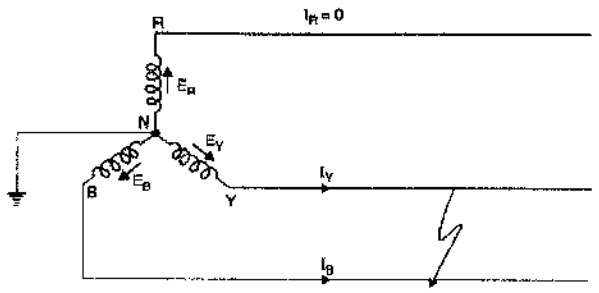
(10 Marks)

03.

a) Explain the different types of series reactors which are used in power system with suitable sketch diagram?

(10 Marks)

b) Consider Line to Line fault as shown in the figure.



i. Draw the sequence impedance Diagram

ii. Show that fault current is equal to the value of $\frac{j\sqrt{3}E_R}{Z_1+Z_2}$

(Z_1 and Z_2 are positive and negative sequence impedance. State any assumption you may take.)

(10 Marks)

4.

a) Define the term relative to the Relay

i. Pick-up current

ii. Current setting

iii. Plug-setting multiplier (P.S.M.).

iv. Time setting

v. Time/P.S.M. Curve

(10 Marks)

b) Name the fig Q4 relay and explain the operation of the relay.

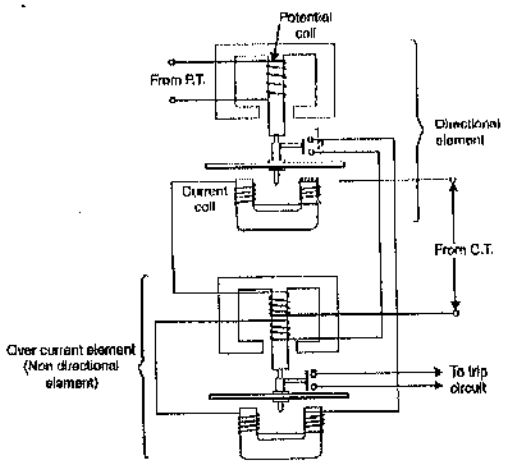


Fig Q4 (10 Marks)

05.

a). Explain the operation of the Buchholz relay for minor fault and major fault of the transformer with suitable sketch.

(10 Marks)

b). Merz-Price circulating-current scheme for the protection of a 3-phase in star/delta power transformer of 220/11,000 line volts is connected against phase to ground and phase-to-phase faults. The protective transformers on 220 V side have a current ratio of 600/5 of the current transformers. What should be the current ratio of current transformer on 11,000 V side.

(10 Marks)

06.

a) Name five causes for the Motor failure and Explain protection methods for each cause.

(10 Mark)

b) The basic method for busbar protection is the differential protection scheme and basic method for Line protection is Distance protection.

i. Explain the operation of the differential protection scheme for the protection of the bus bar with suitable sketches. (5 Marks)

ii. Explain the operation of the Distance protection scheme for the protection of line with suitable sketches. (5 Marks)